

ALEX HEMBROM

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EDUCATION

Government College of Engineering and Leather Technology

B.Tech in Computer Science and Engineering | CGPA: 8.19

Anandamath Vidyapith - Class-XII(Science)

West Bengal Council of Higher Secondary Education

Ramakrishna Vivekananda Mission Vidyapith - Class-X

West Bengal Board of Secondary Education

Kolkata, West Bengal

June 2021 - May 2025

Ichapur, West Bengal

Barrackpore, West Bengal

SKILLS

- **Technical Skills:** Data Structures & Algorithms, Web Development.
- **Programming Languages :** Java, Python, C, JavaScript.
- **Frontend:** HTML, CSS, React.js, React Native.
- **Backend:** Node.js.
- **Database:** Firebase, MySQL.
- **Developer Tools:** Git, GitHub, VS Code.
- **AI Productivity Tools:** Kiro, GitHub Copilot, OpenAI Codex, Claude, Gemini, ChatGPT, Antigravity AI.

PROJECTS

CRYPTO WALLET - [\[github\]](#)

- Developed a **mobile cryptocurrency wallet** using **React Native and Firebase** with secure user authentication and real-time data synchronization.
- Integrated **Web3 functionality** using **Ethers.js** and **Ganache** to create **Ethereum wallets**, **track live balances**, and execute ETH transactions.
- Designed a responsive **neon-style UI** with **dynamic asset views**, **smooth navigation**, and **interactive components**.
- Leveraged Kiro for specification-driven development and GitHub Copilot, OpenAI Codex, and Antigravity AI to accelerate development, debugging, and code optimization.
- **Tech Used:** React Native, Expo, Firebase, Ethers.js, JavaScript, Ganache.

AGENTIC HONEYPOT - (Collaborative Project) [\[github\]](#)

- Developed an **AI-powered system** using **FastAPI** to detect **financial scam messages** and simulate interactions with scammers.
- Contributed to building **multi-turn conversation logic** with **intelligent responses** to extract details like **bank accounts** and **UPI IDs**.
- Implemented **keyword scoring** and **regex-based pattern** matching to **identify malicious intent** and extract **phishing** data.
- **Tech Used:** Python, FastAPI, Regex, REST APIs, Uvicorn.

SUTRA (Gamified NGO Assistant) - [\[Live demo\]](#)

- Implemented a **gamified React application** to help **NGOs automate program design** using the **Logical Framework Approach (LFA)**.
- Built a **multi-level wizard** with **Firebase Firestore** for **real-time cloud storage** and efficient **state management**.
- Designed a **responsive cyberpunk-themed UI** using **Tailwind CSS**, featuring **dynamic progress tracking** and engaging visuals.
- **Tech Used:** React.js, Firebase, Tailwind CSS, Vite.

TRAFFIC SIGN RECOGNITION SYSTEM - [\[github\]](#)

- Developed a **machine learning desktop application** to identify **43 types of traffic signs**.
- Trained a **custom CNN model** using **Keras** and **OpenCV**, and integrated it with a **Tkinter-based GUI** for easy image testing.
- Enabled **real-time image processing** to accurately predict and display the correct traffic sign from user-uploaded images.
- **Tech Used:** Python, Keras, OpenCV, Tkinter.

ACHIEVEMENTS

- **Google Cloud Skills Boost:** Participated in Google Cloud Skills Boost program, earning 50+ badges, including "Build, Train and Deploy ML Models with Keras on Google Cloud." [Google Cloud Badge](#)
- **Hackathons:** Participated in the nationwide **GUVI HCL Hackathon** to build and deploy technical solutions. [Certificate](#)
- **Core member of the Web Development and Photography Team at GDG GCELT**, responsible for building web solutions and capturing event highlights.